ID & Name: 150721854 -Yunus Emre Turunç
150720018 - Semiha Şen

Pre-LAB Work

1. For the circuits given in Fig. 1, calculate the low (f_L) and high (f_H) cut-off frequencies. Assume β=200, Cμ=3pF, Cπ=15pF. (Vcc=15V and Vee=-15V)

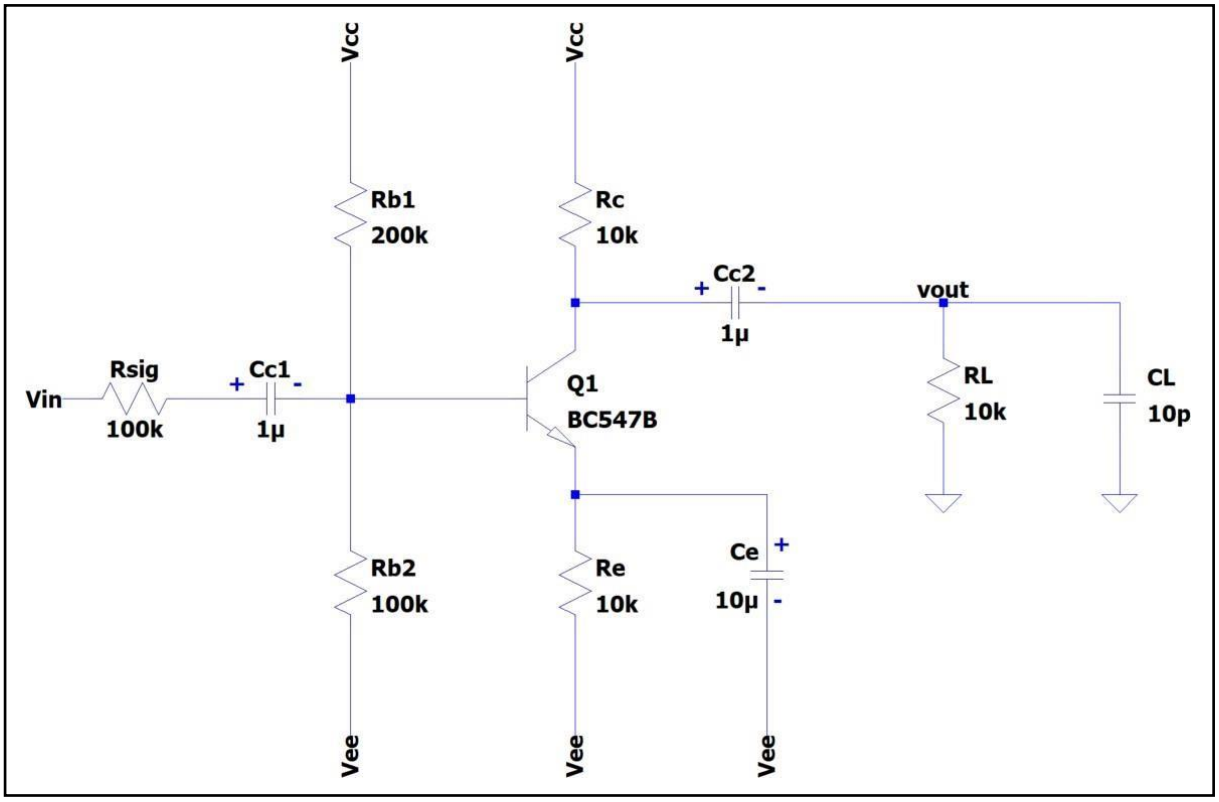
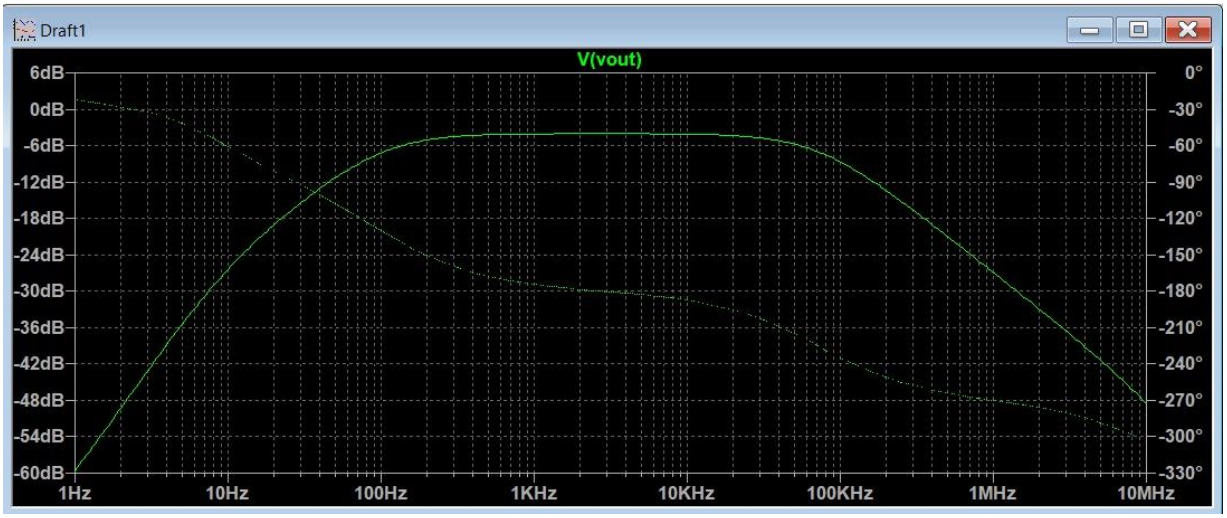


Figure 1

Answer: F low = 105,7 (Hz) F high = 88K (Hz)

2. Simulate the circuit in Fig. 1 with SPICE. Let Vcc=15V and Vee=-15V. Conduct an AC analysis using the following spice command and plot the amplitude of the frequency response. Indicate the low (f_L) and high (f_H) cut-off frequencies.

   .ac dec 10 1 10e6 F low = 103(Hz) F high = 72K (Hz)



3. For the circuits given in Fig. 2, calculate the low (f_L) and high (f_H) cut-off frequencies. Assume β=200, Cμ=3pF, Cπ=15pF. (Vcc=15V and Vee=-15V)

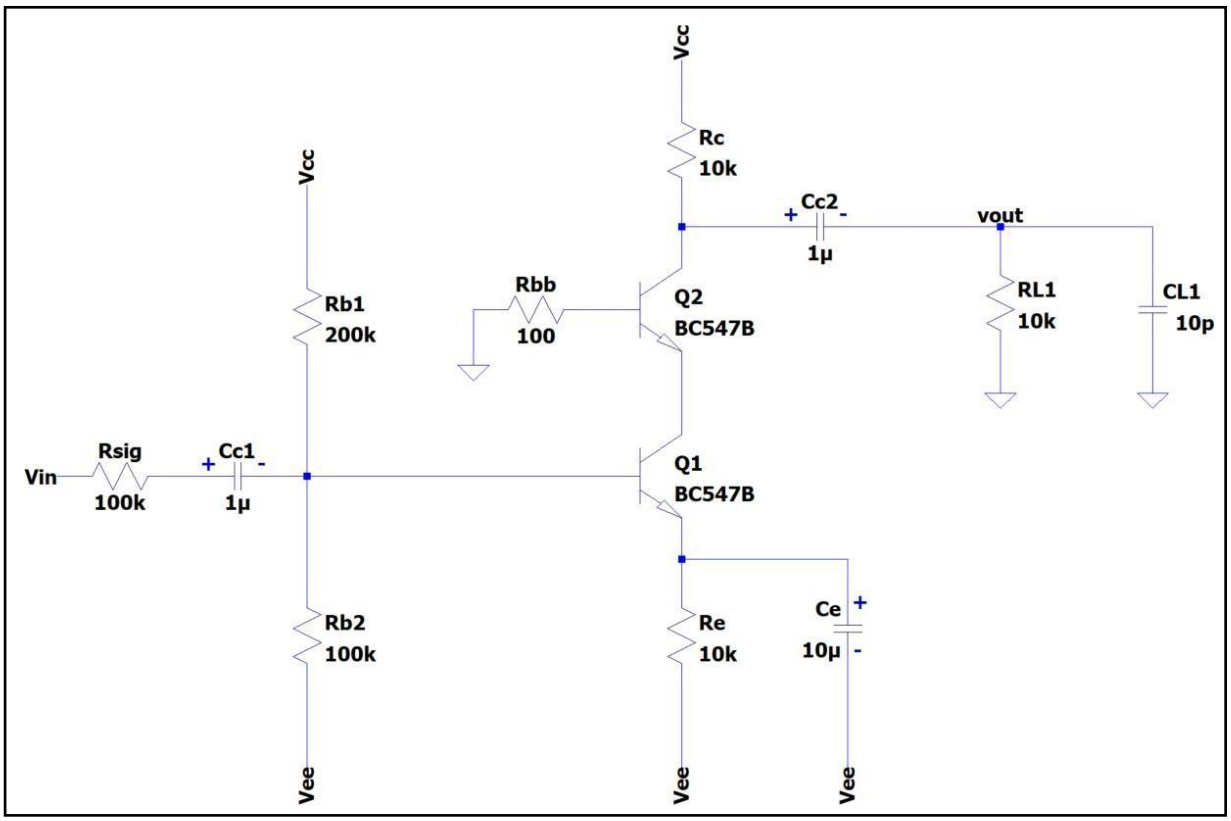
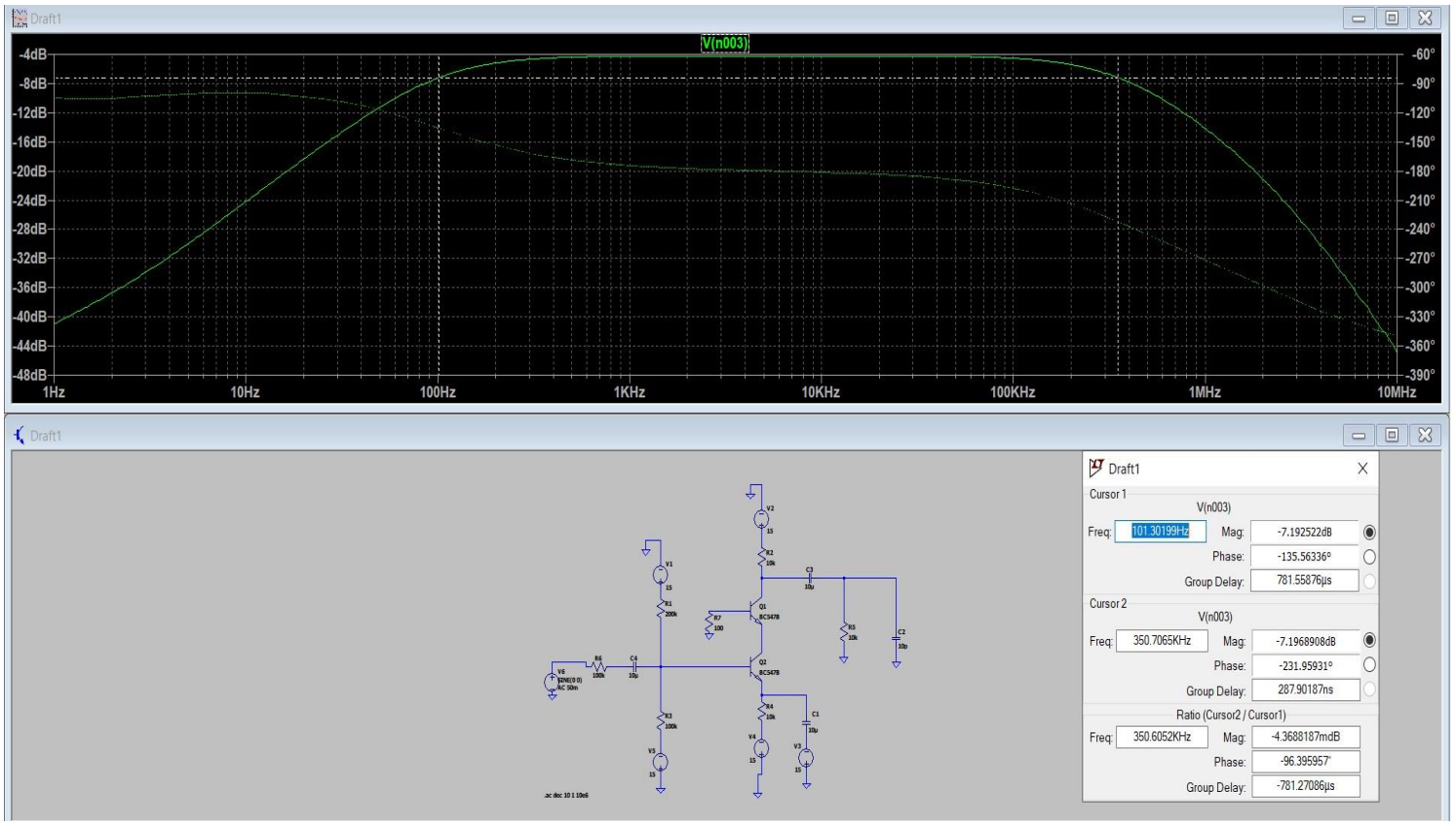


Figure 2

Answer: F low = 100 (Hz)
F high = 340K (Hz)

4. Simulate the circuit in Fig. 2 with SPICE. Let Vcc=15V and Vee=-15V. Conduct an AC analysis and plot the amplitude of the frequency response. Indicate the low (f_L) and high (f_H) cut-off frequencies. F low = 101(Hz) F high = 350K (Hz)



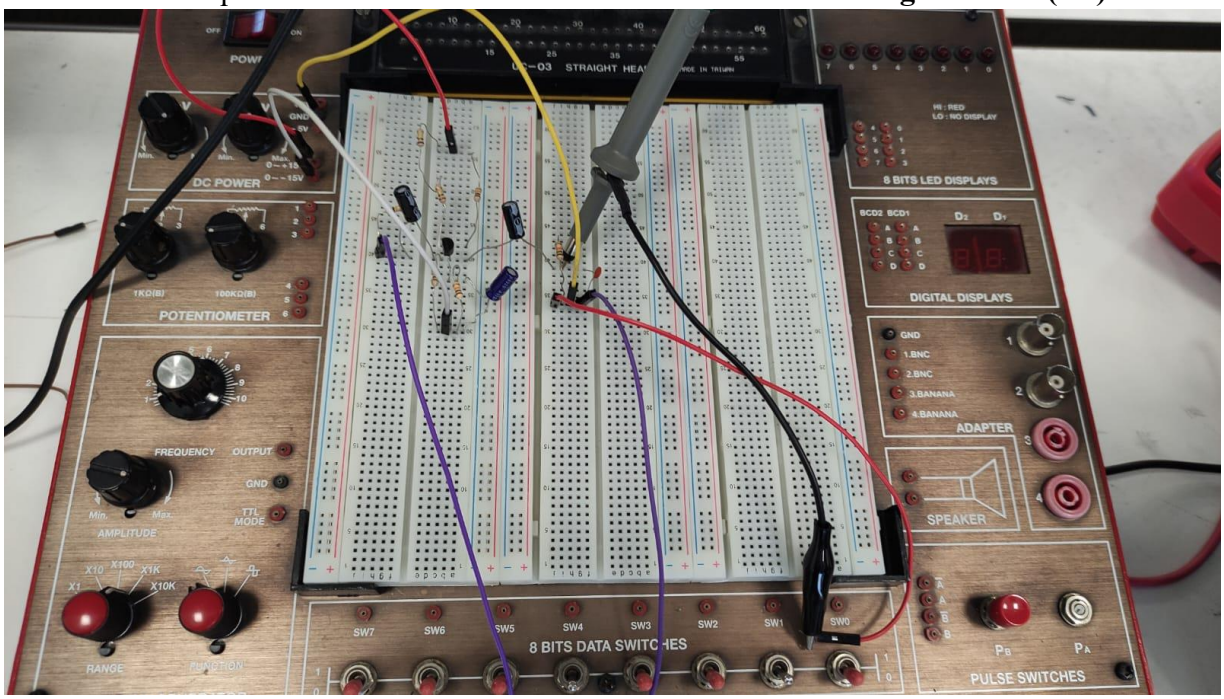
Equipment/Component List

Table 1: Equipment/Component List

Equipment/Component	Model/Value	Count
Breadboard	-	1
Resistor	100Ω	1
Resistor	10kΩ	**3**
Resistor	100kΩ	**2**
Resistor	200kΩ	1
Capacitor	10pF	1
Capacitor	1µF	2
Capacitor	10µF	1
BJT (npn)	BC547	**2**
Multi-meter	-	1
Function Generator	-	1
Oscilloscope	-	1
Voltage Source	15V, -15V	1
Probes & Cables	-	-

Experiment Procedure

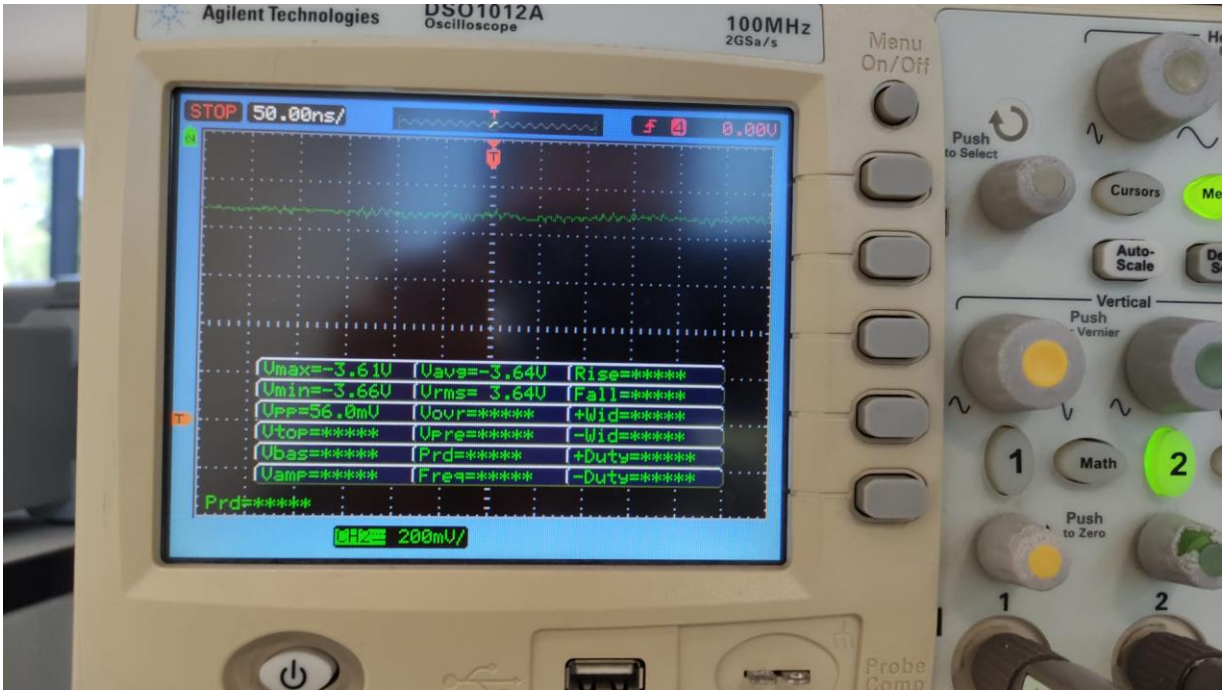
5. Adjust Vcc to 15V and Vee to -15V. You will also need the ground connection. Do not connect the supplies.
6. Set-up the circuit in Fig. 1. Be very careful while connecting the capacitor. The +/- pins must be connected as shown in the Fig. 1.
7. Connect the ground and then Vee and Vcc.
8. Apply sine waves of 10Hz, 100Hz, 1kHz, 10kHz, 100kHz, 1MHz, 10MHz with amplitudes of 50mV to Vin.
9. Also find the 3-dB cut-off frequencies.
10. Plot the gain in dB as a function of the frequency. Indicate the low (f_L) and high (f_H) cut-off frequencies. **F low ~ 300 Hz** **F high ~10.1K (Hz)**

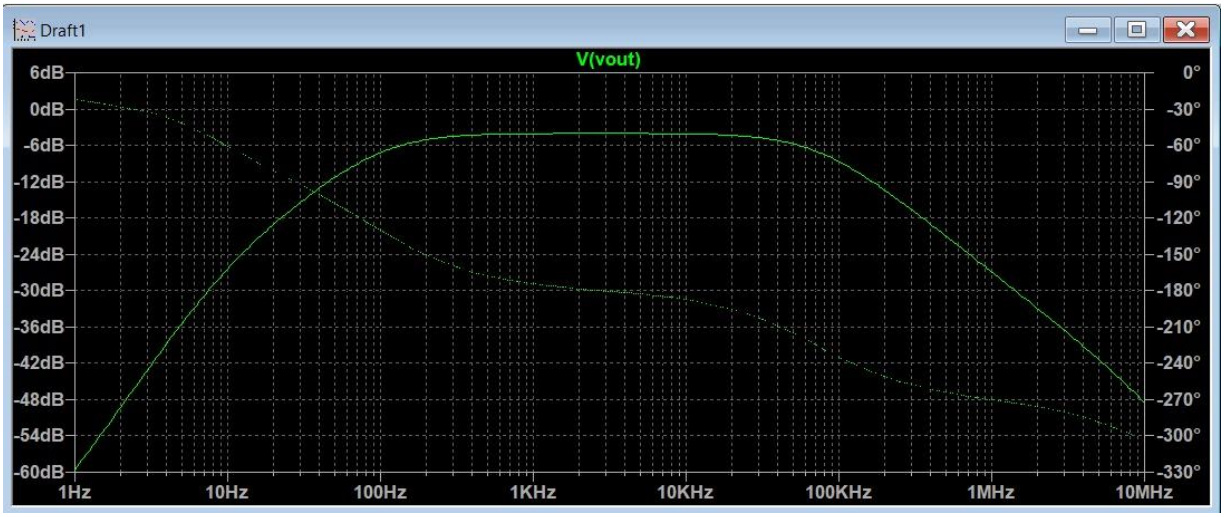


Lab and LTSpice Frequency Response Results

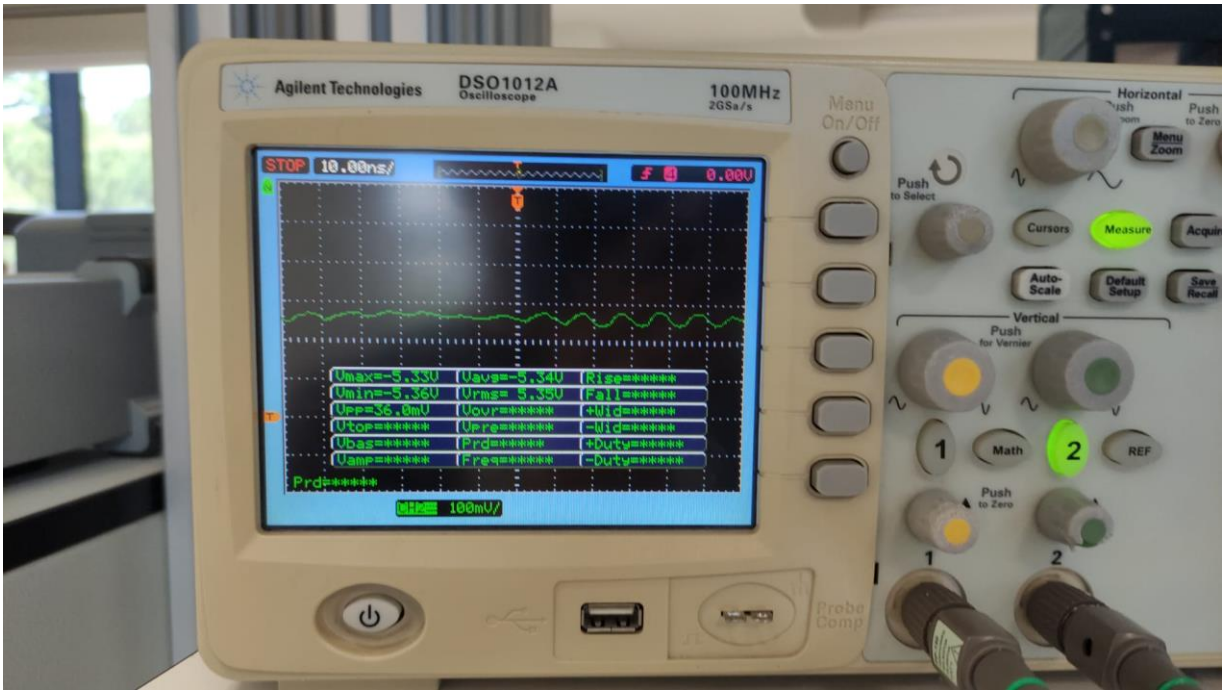
1.Devre

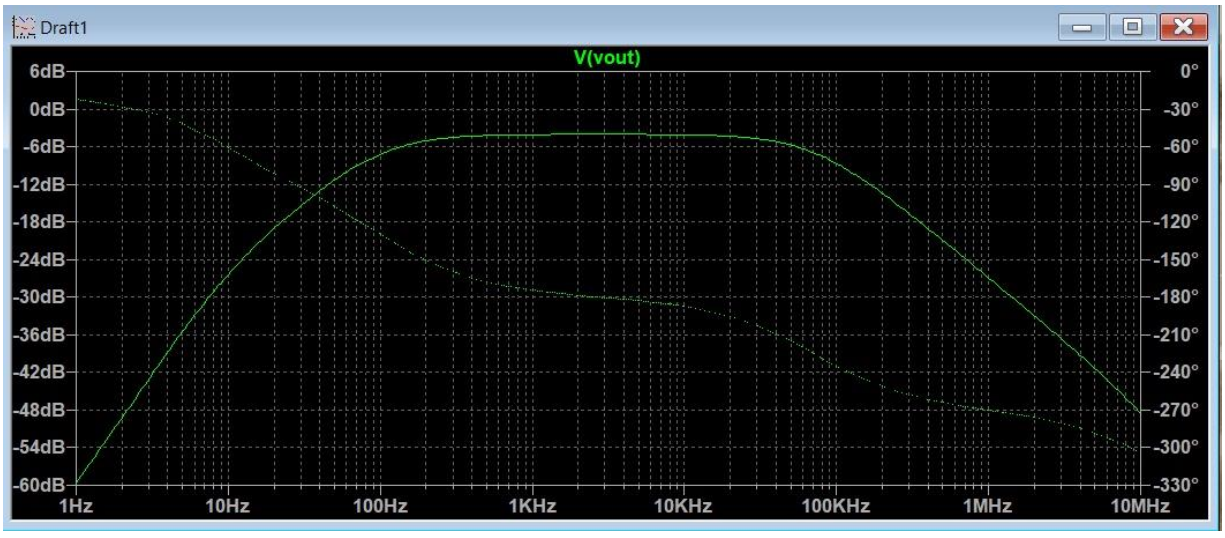
10HZ




Draft1
V(vout)
6dB
0dB
-6dB
-12dB
-18dB
-24dB
-30dB
-36dB
-42dB
-48dB
-54dB
-60dB
0°
-30°
-60°
-90°
-120°
-150°
-180°
-210°
-240°
-270°
-300°
-330°
1Hz
10Hz
100Hz
1KHz
10KHz
100KHz
1MHz
10MHz

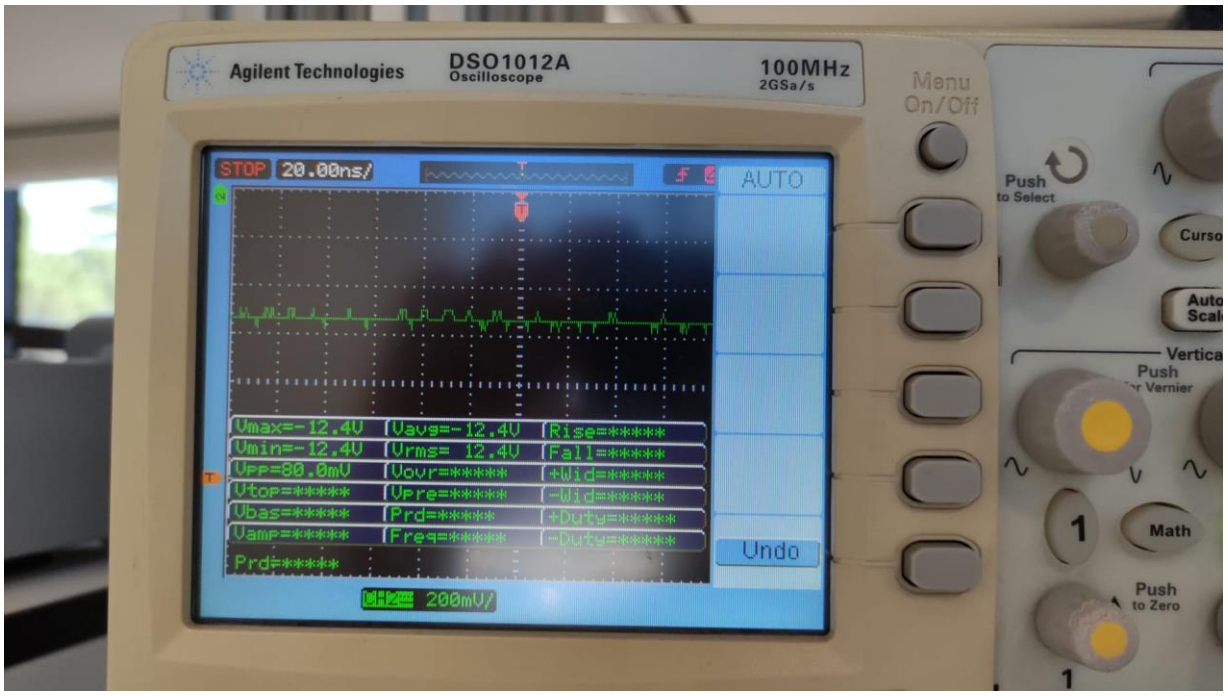
100Hz

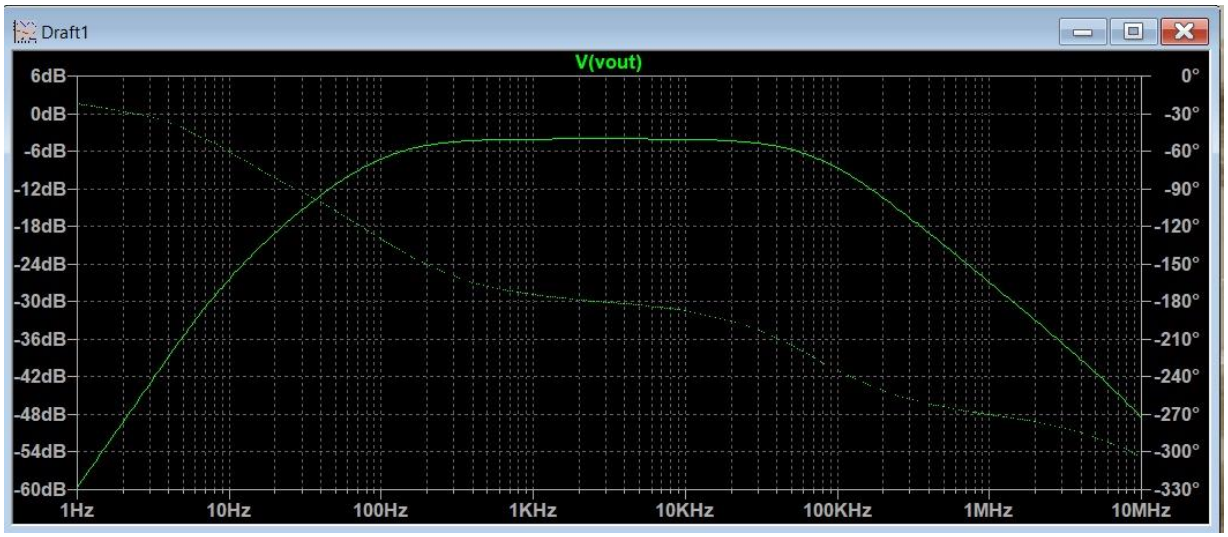

Agilent Technologies
DSO1012A
Oscilloscope
100MHz
2GSa/s
Menu On/Off
Horizontal
Menu Zoom
Push to Select
Cursors
Measure
Auto-Scale
Default Setup
Vertical
Math
REF
STOP
10.00ns/
0.00V
Vmax=-5.33V
Vmin=-5.36V
Vpp=36.0mV
Vtop=
Vbas=
Vamp=
Vavg=-5.34V
Vrms= 5.35V
Vovr=
Vpre=
Prd=
Freq=
Rise=
Fall=
+Wid=
-Wid=
+Duty=
-Duty=
100mV/


Draft1
V(vout)
6dB
0dB
-6dB
-12dB
-18dB
-24dB
-30dB
-36dB
-42dB
-48dB
-54dB
-60dB
0°
-30°
-60°
-90°
-120°
-150°
-180°
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1Hz
10Hz
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1KHz
10KHz
100KHz
1MHz
10MHz

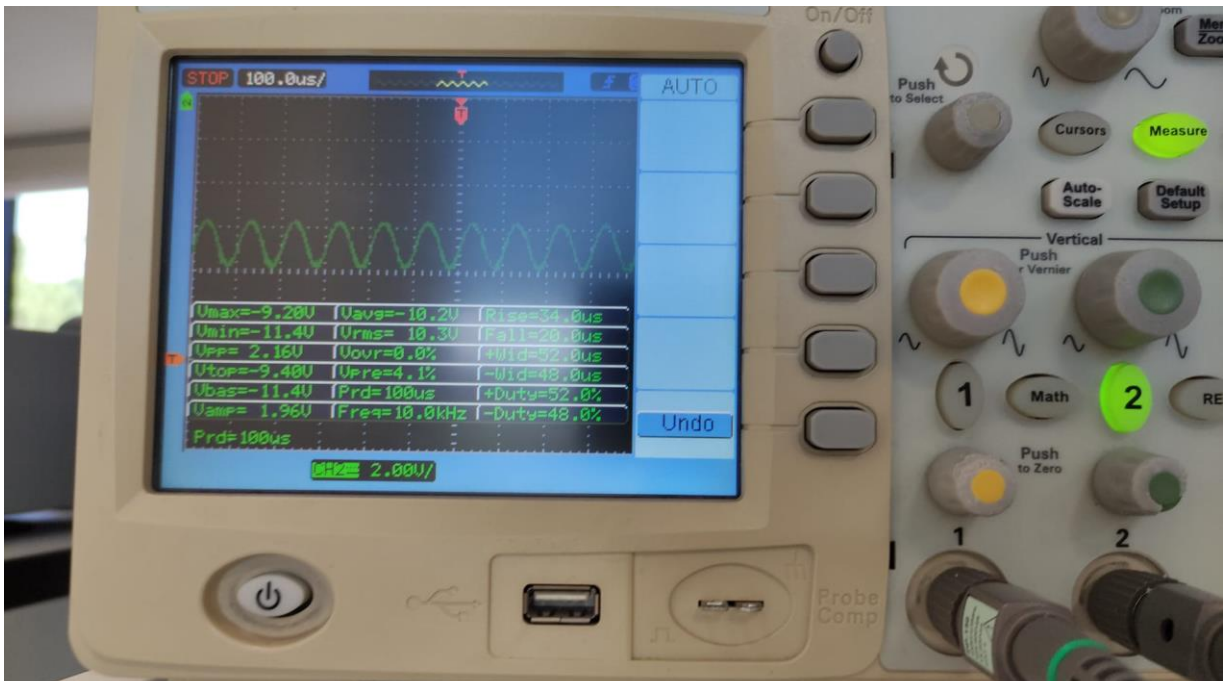
Freq. Response of Amplifiers Experiment 4

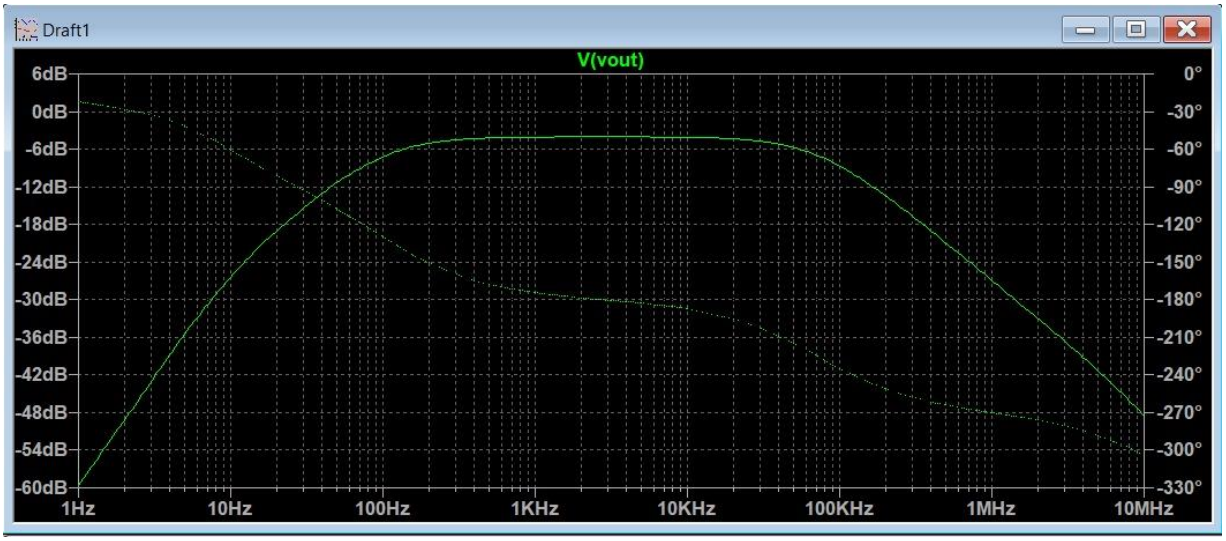
1k Hz



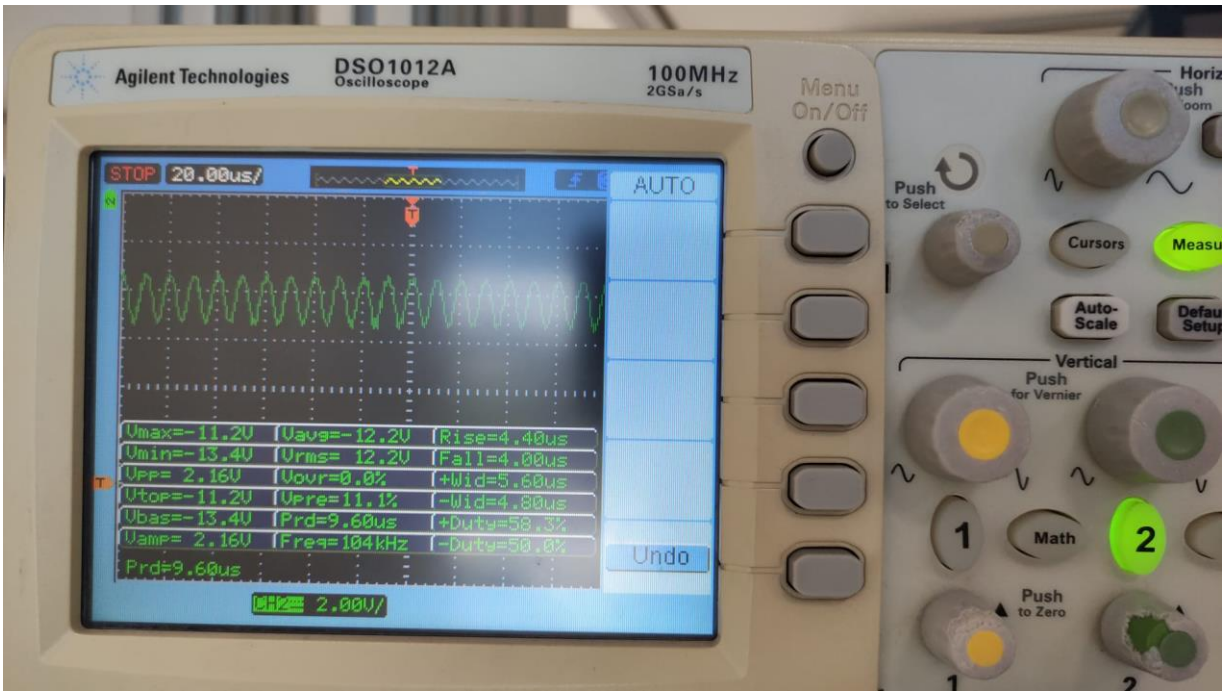


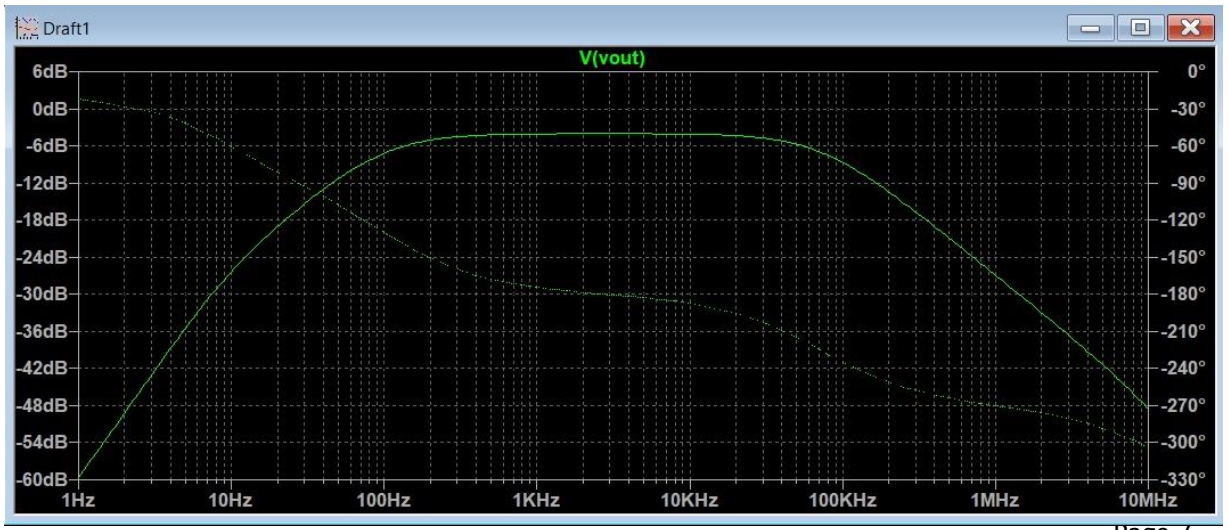
10k Hz




Draft1
V(vout)
6dB
0dB
-6dB
-12dB
-18dB
-24dB
-30dB
-36dB
-42dB
-48dB
-54dB
-60dB
0°
-30°
-60°
-90°
-120°
-150°
-180°
-210°
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-330°
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10Hz
100Hz
1KHz
10KHz
100KHz
1MHz
10MHz

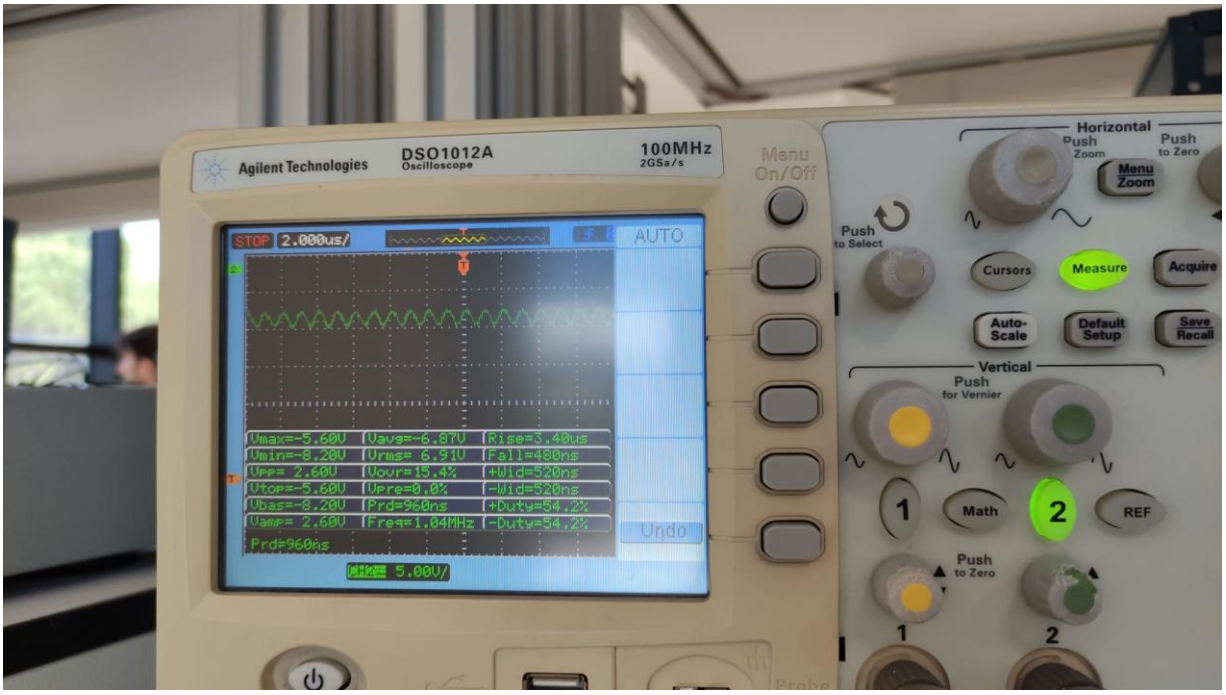
100k Hz

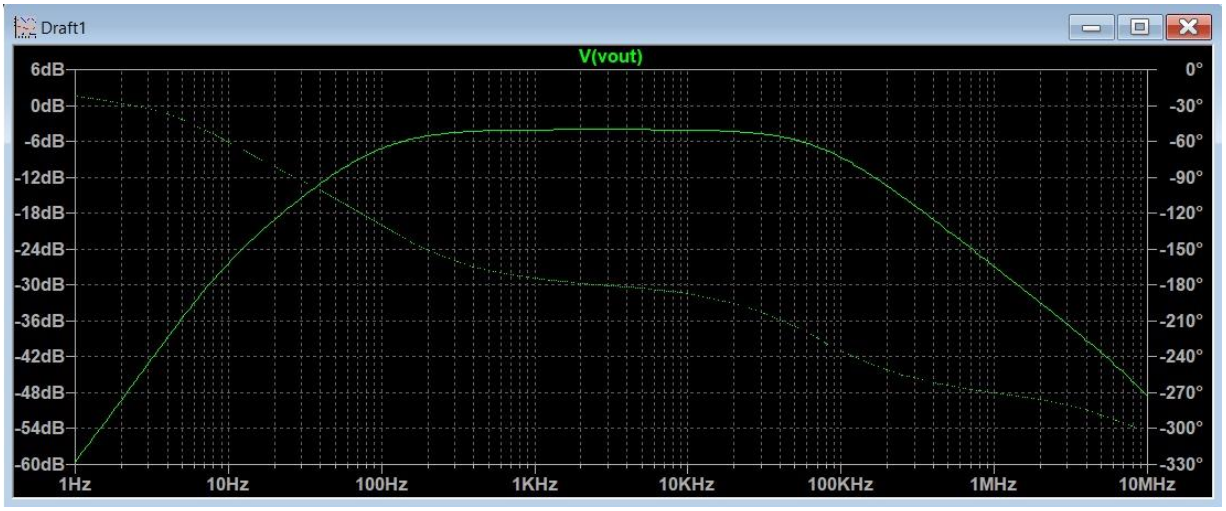

Agilent Technologies
DSO1012A
Oscilloscope
100MHz
2GSa/s
Menu
On/Off
STOP
20.00us/
AUTO
Vmax=-11.2V
Vmin=-13.4V
Vpp= 2.16V
Vtop=-11.2V
Vbas=-13.4V
Vamp= 2.16V
Vavg=-12.2V
Vrms= 12.2V
Vovr=0.0%
Vpre=11.1%
Prd=9.60us
Freq=104kHz
Rise=4.40us
Fall=4.00us
+Wid=5.60us
-Wid=4.80us
+Duty=58.3%
-Duty=50.0%
Prd=9.60us
Undo
2.00V/
Push
to Select
Cursors
Auto-
Scale
Vertical
Push
for Vernier
1
Math
2
Push
to Zero


Draft1
V(vout)
6dB
0dB
-6dB
-12dB
-18dB
-24dB
-30dB
-36dB
-42dB
-48dB
-54dB
-60dB
0°
-30°
-60°
-90°
-120°
-150°
-180°
-210°
-240°
-270°
-300°
-330°
1Hz
10Hz
100Hz
1KHz
10KHz
100KHz
1MHz
10MHz

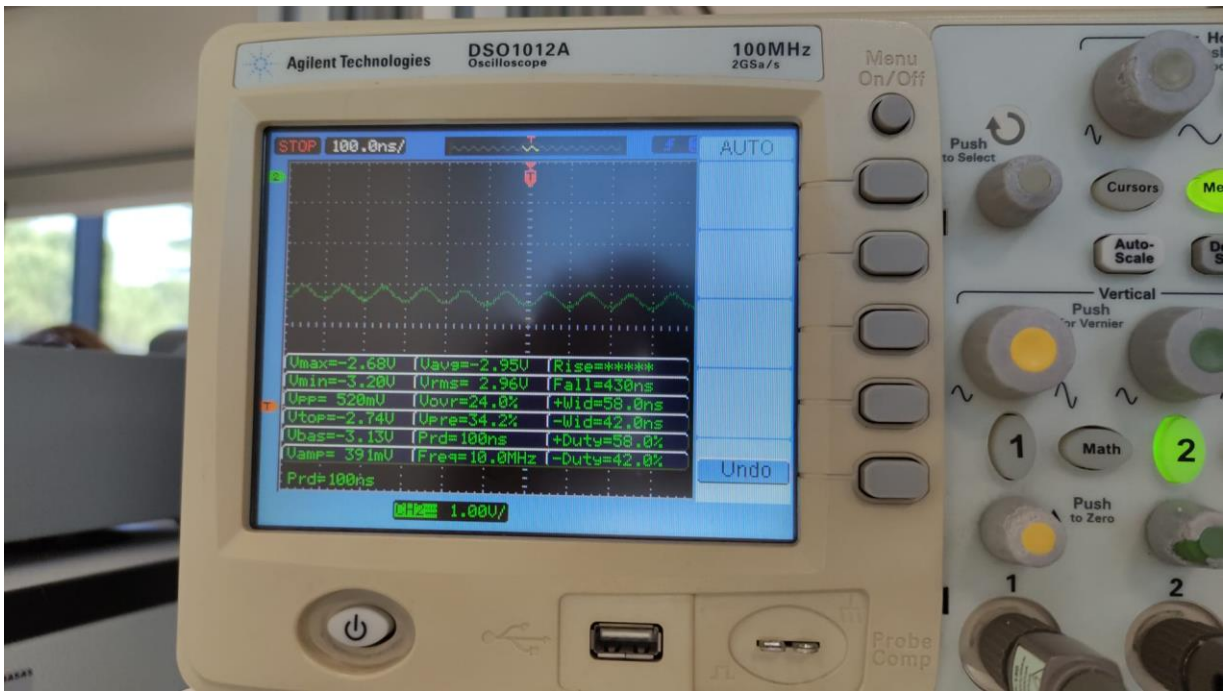
Freq. Response of Amplifiers Experiment 4

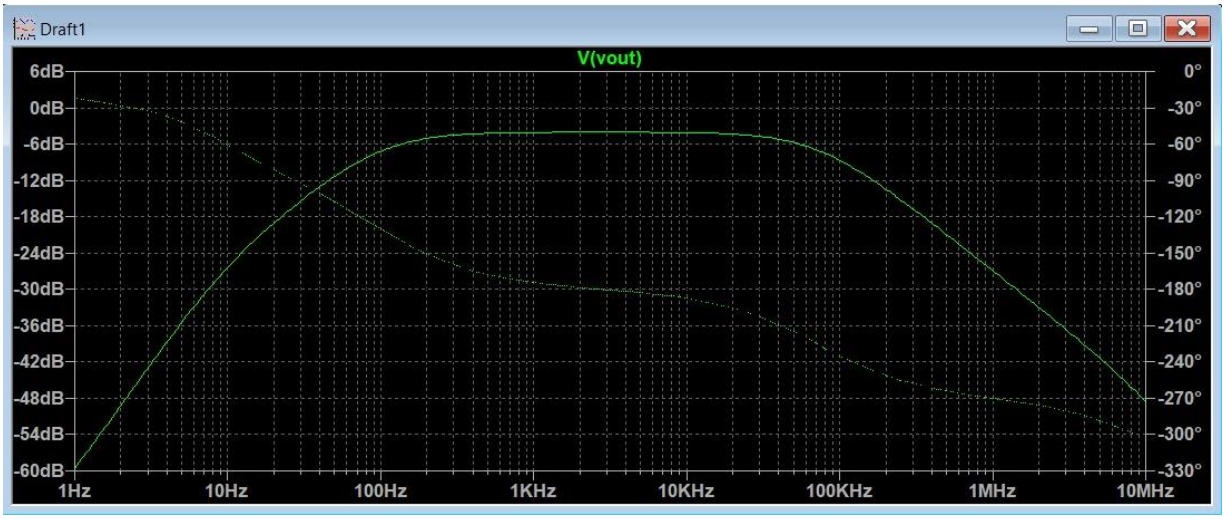
1M Hz





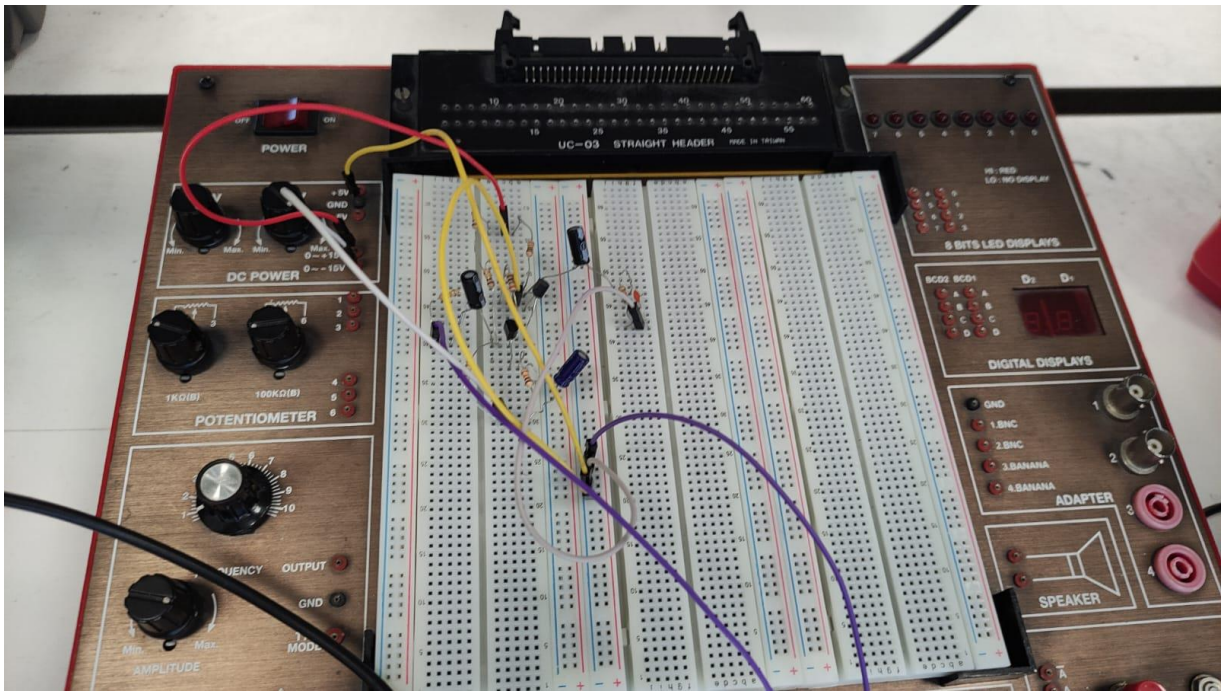
10M Hz





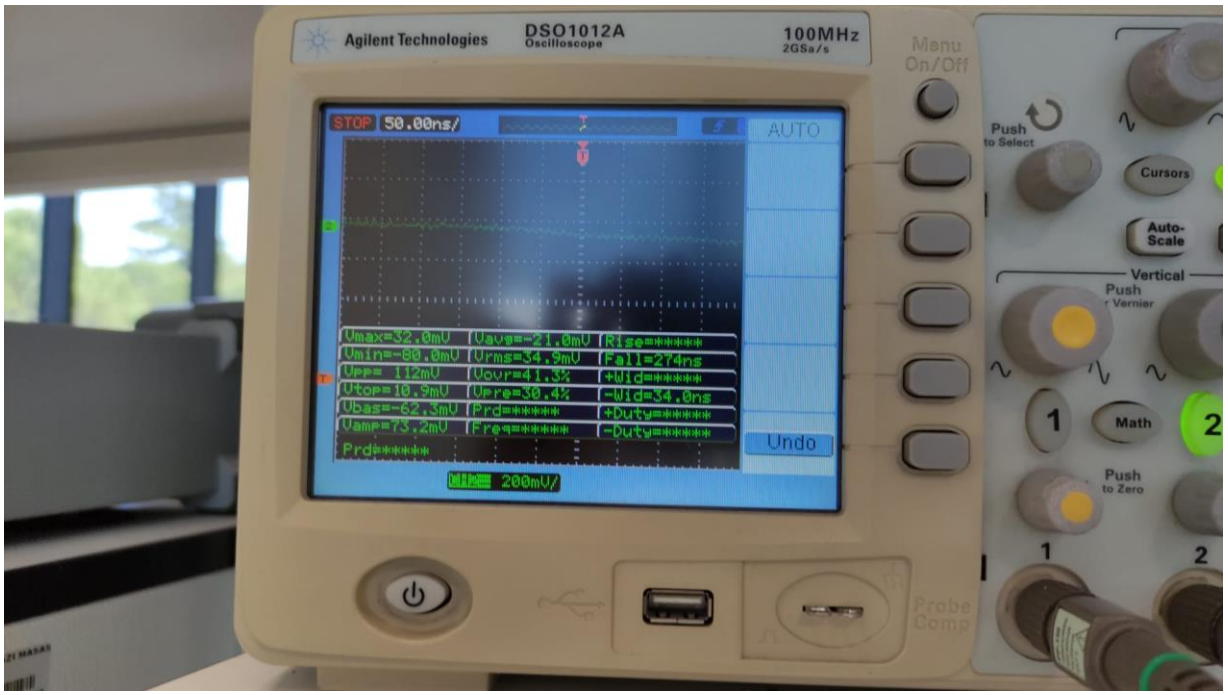
12. Power down and disassemble the circuit.
13. Set-up the circuit in Fig. 2. Be very careful while connecting the capacitor. The +/- pins must be connected as shown in the Fig. 2.
14. Connect the ground and then Vee and Vcc. Power up your circuit.
15. Apply sine waves of 10Hz, 100Hz, 1kHz, 10kHz, 100kHz, 1MHz, 10MHz with amplitudes of 50mV to Vin.
16. Also find the 3-dB cut-off frequencies.
17. Plot the gain in dB as a function of the frequency. Indicate the low (f_L) and high (f_H) cut-off frequencies. **Hard to find cut-off frequencies because of derivation**

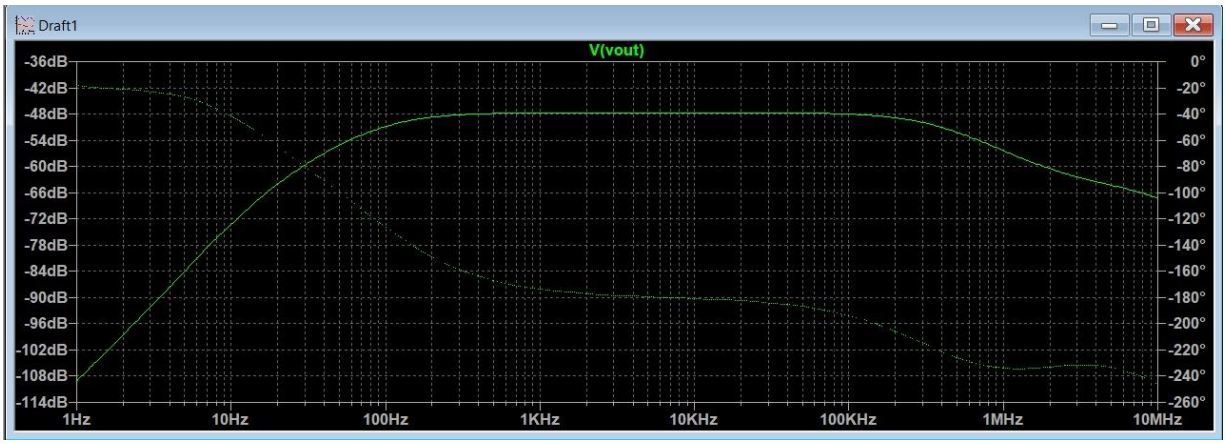
2.Devre



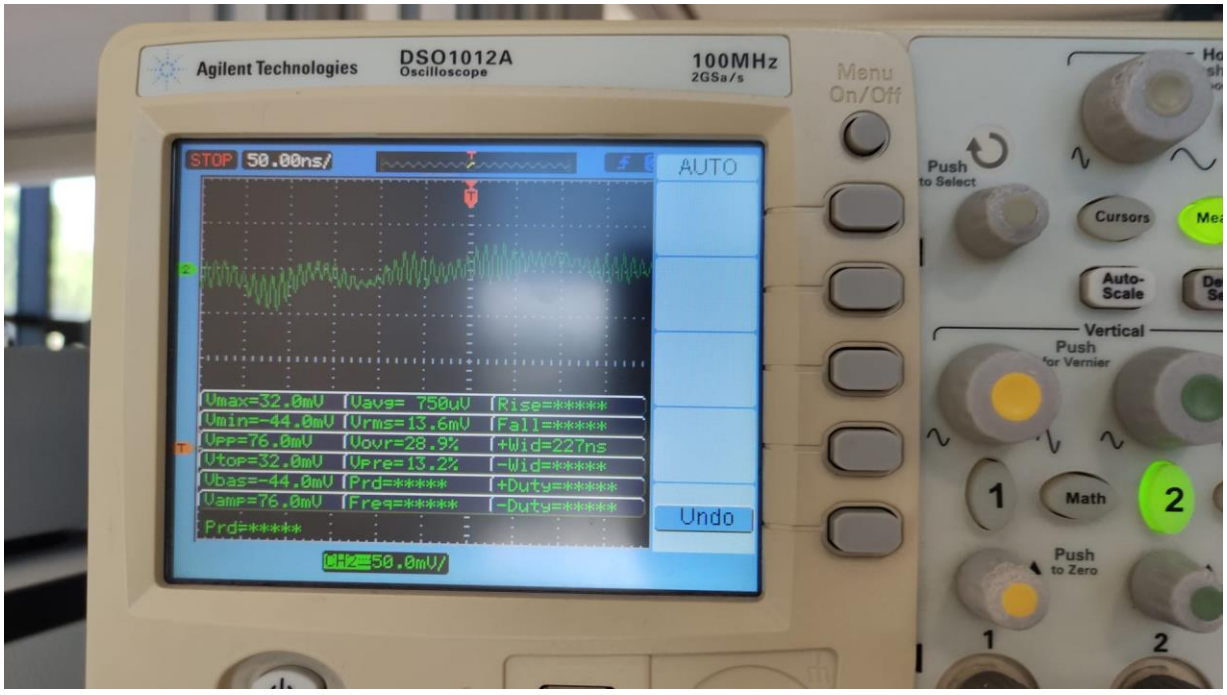
Freq. Response of Amplifiers Experiment 4

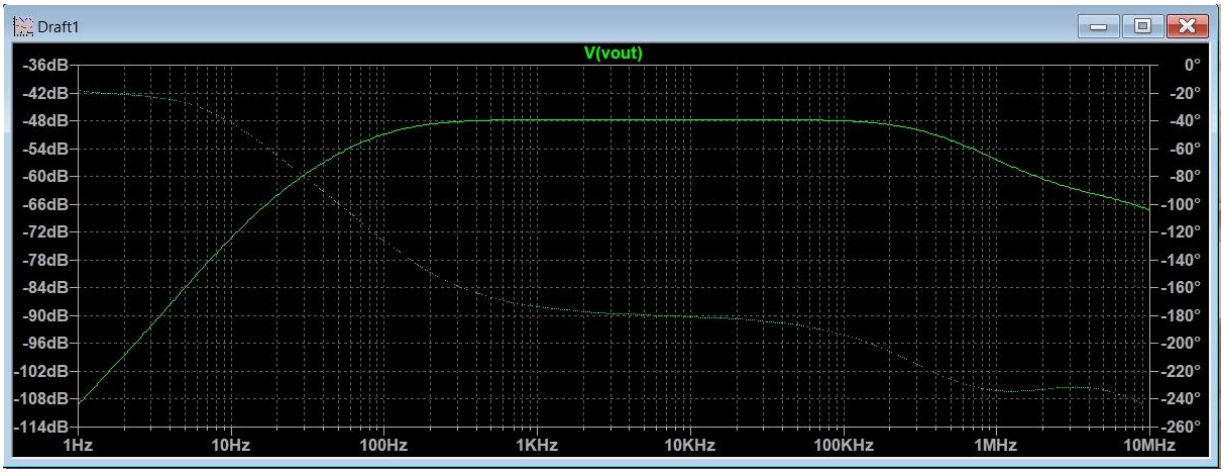
10Hz



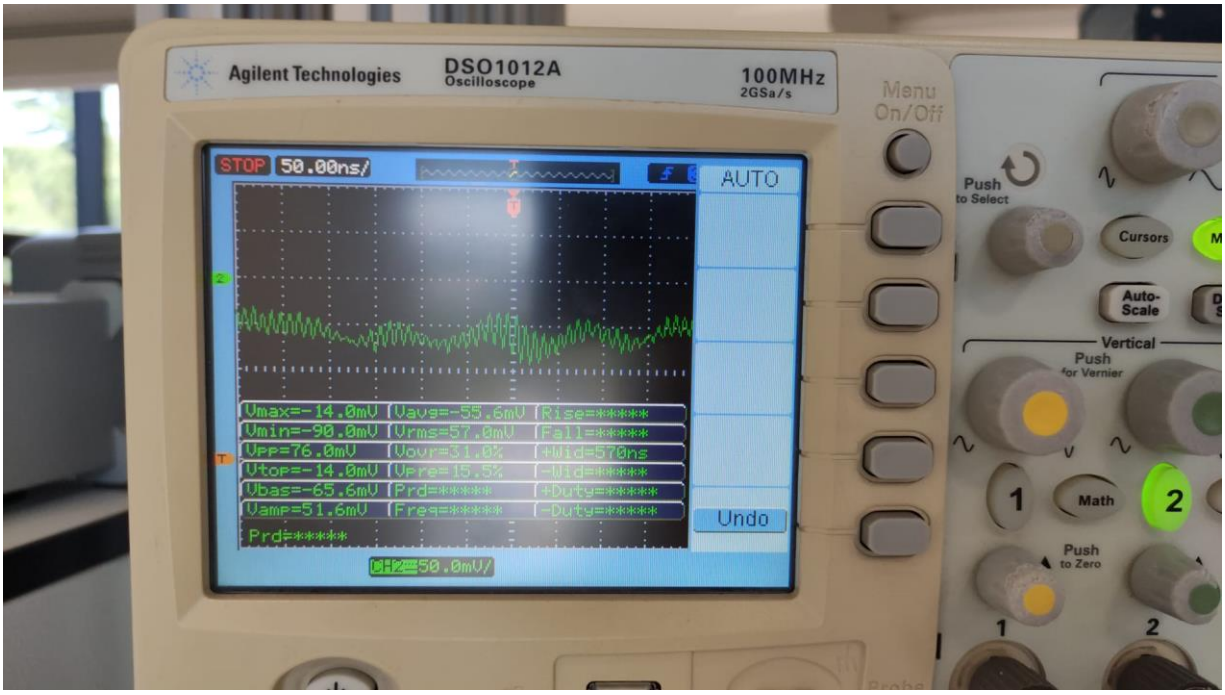


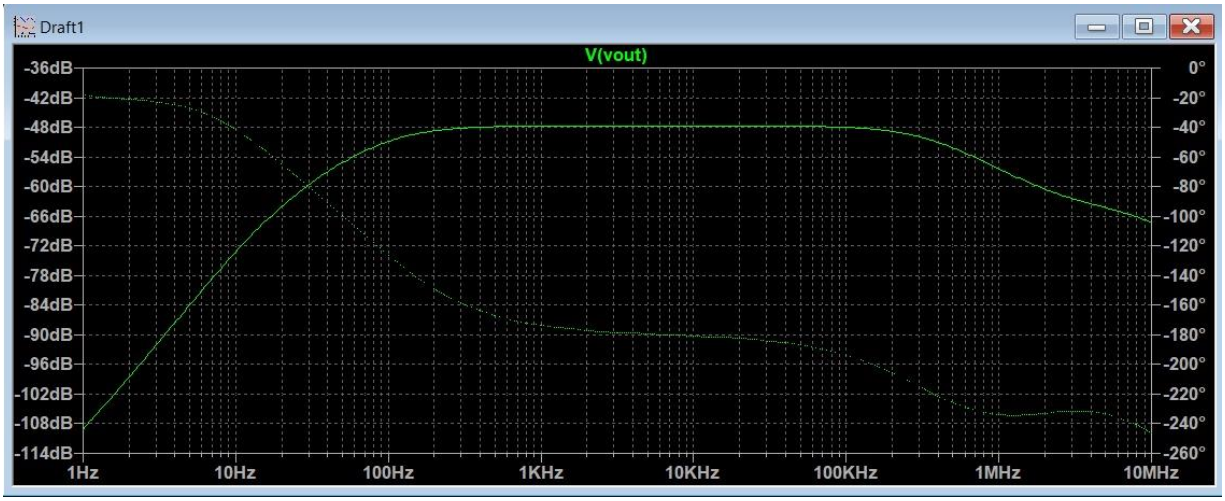
100 Hz




Draft1
V(vout)
-36dB
-42dB
-48dB
-54dB
-60dB
-66dB
-72dB
-78dB
-84dB
-90dB
-96dB
-102dB
-108dB
-114dB
0°
-20°
-40°
-60°
-80°
-100°
-120°
-140°
-160°
-180°
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1KHz
10KHz
100KHz
1MHz
10MHz

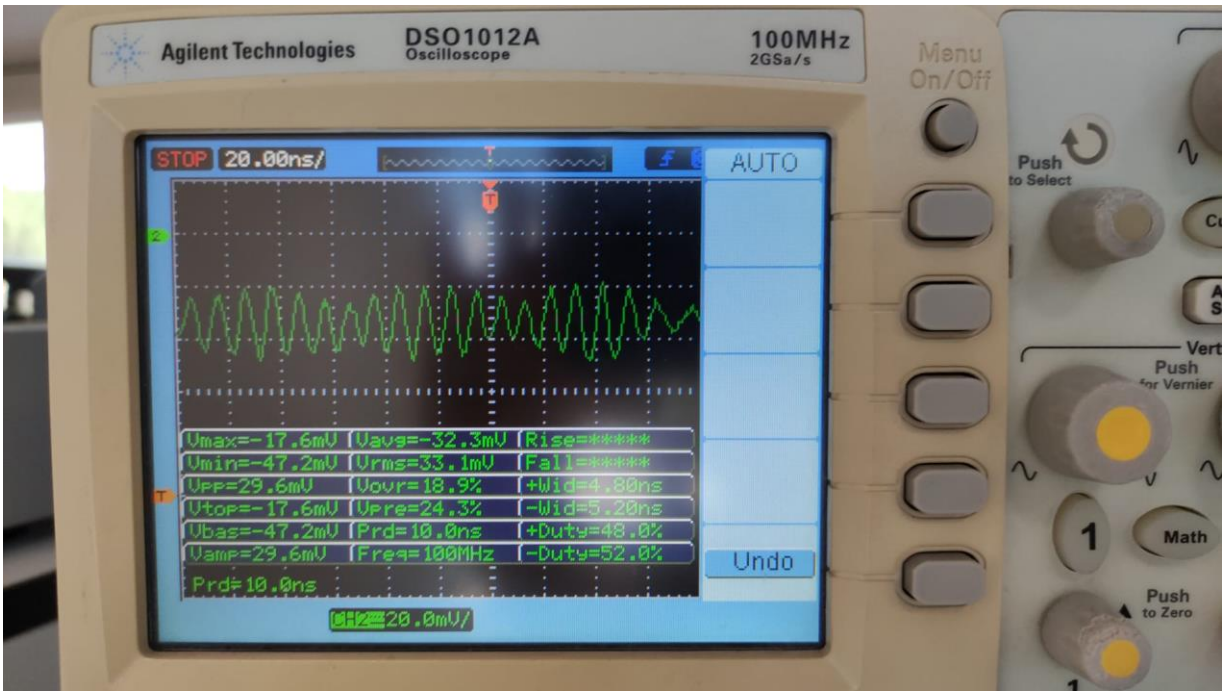
1k Hz

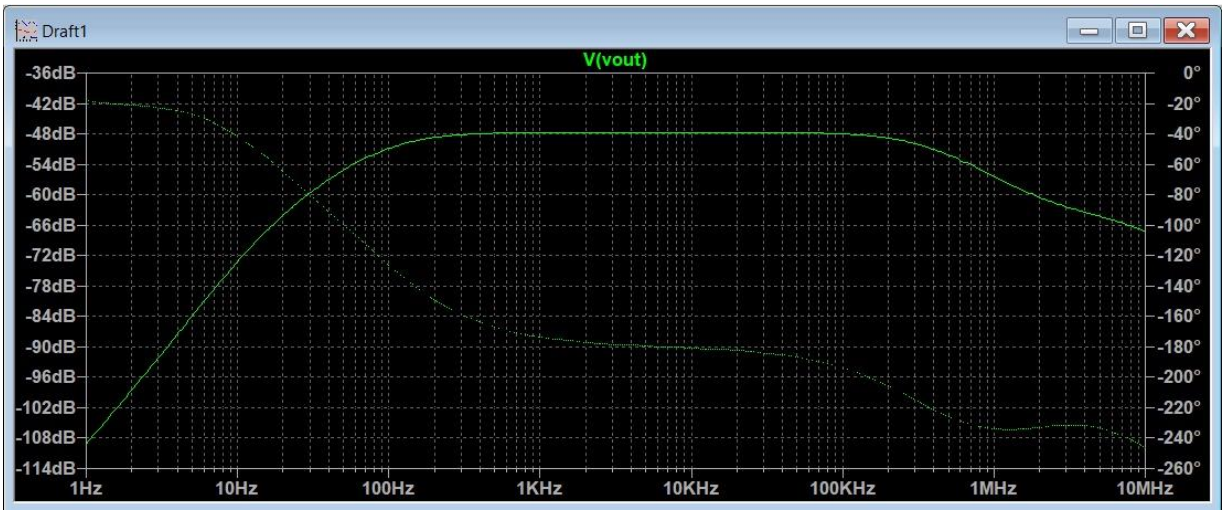

Agilent Technologies
DSO1012A
Oscilloscope
100MHz
2GSa/s
Menu
On/Off
STOP
50.00ns/
AUTO
Vmax=-14.0mV
Vmin=-90.0mV
Vpp=76.0mV
Vtop=-14.0mV
Vbas=-65.6mV
Vamp=51.6mV
Vavg=-55.6mV
Vrms=57.0mV
Vovr=31.0%
Vpre=15.5%
Prd=*****
Freq=*****
Rise=*****
Fall=*****
+Wid=570ns
-Wid=*****
+Duty=*****
-Duty=*****
Prd=*****
Undo
CH2 50.0mV/
Push
to Select
Cursors
Auto-
Scale
Vertical
Push
for Vernier
1
Math
2
Push
to Zero


Draft1
V(vout)
-36dB
-42dB
-48dB
-54dB
-60dB
-66dB
-72dB
-78dB
-84dB
-90dB
-96dB
-102dB
-108dB
-114dB
0°
-20°
-40°
-60°
-80°
-100°
-120°
-140°
-160°
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10Hz
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1KHz
10KHz
100KHz
1MHz
10MHz

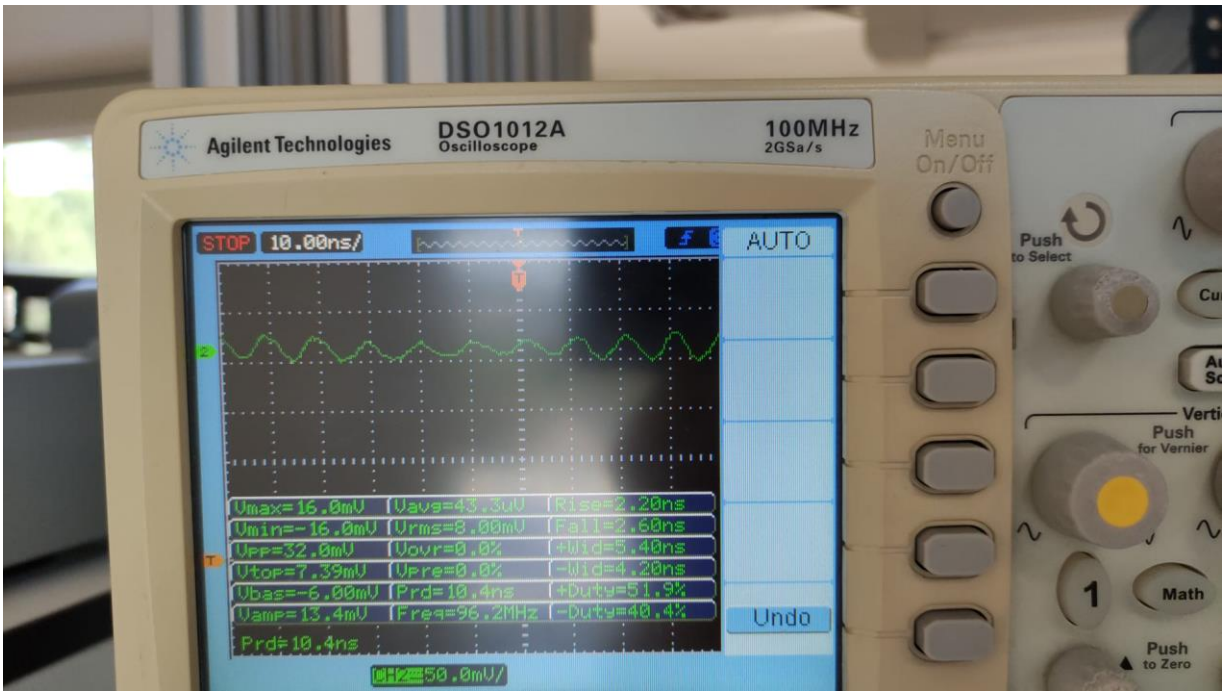
Freq. Response of Amplifiers Experiment 4

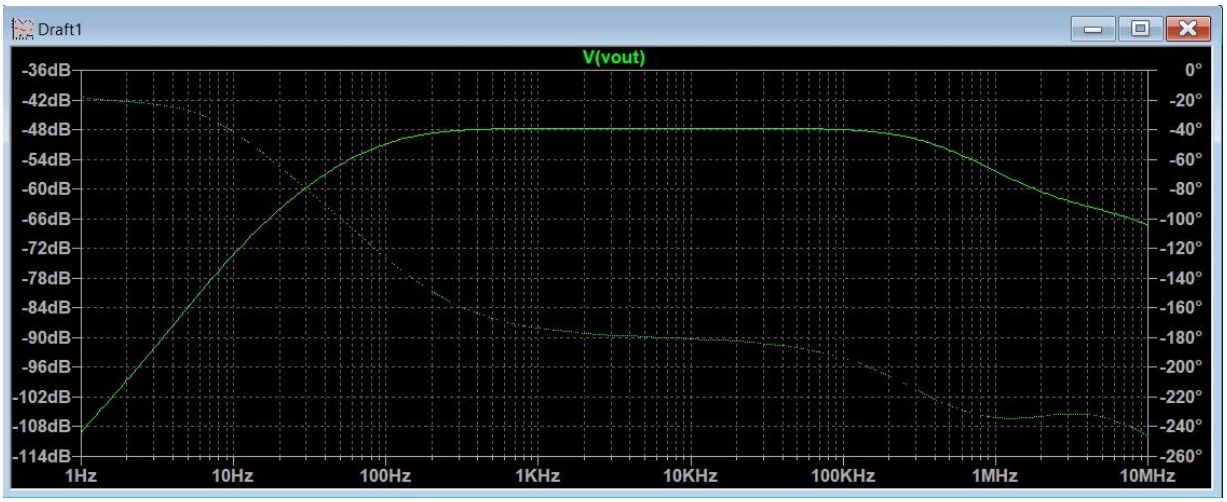
10k Hz



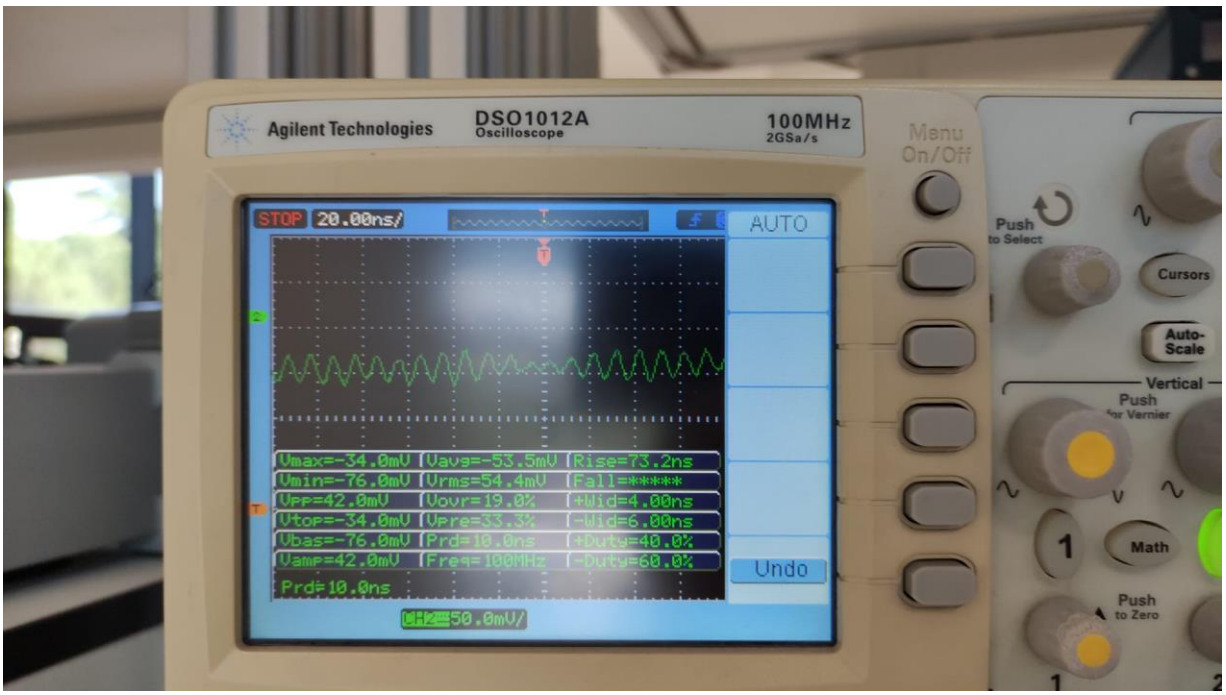


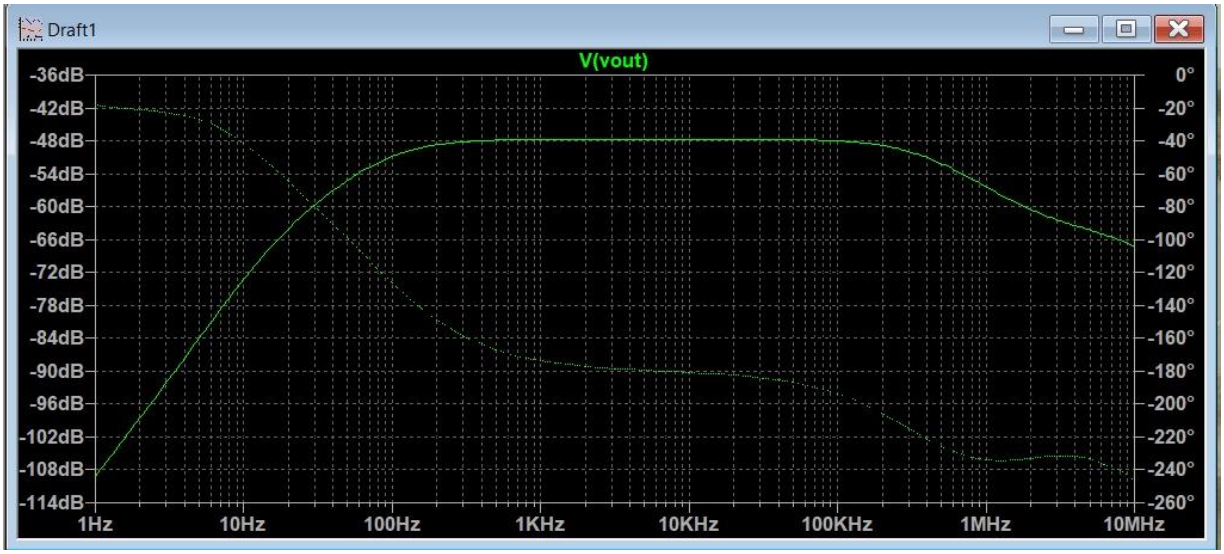
100k Hz



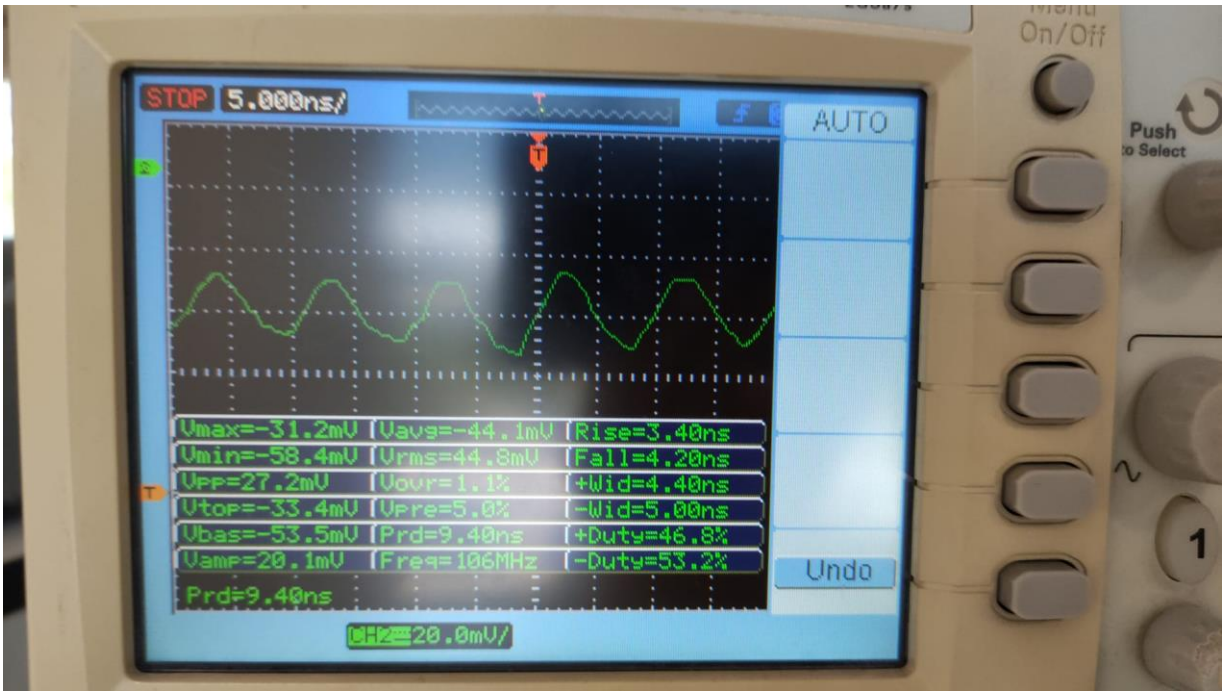

Draft1
V(vout)
-36dB
-42dB
-48dB
-54dB
-60dB
-66dB
-72dB
-78dB
-84dB
-90dB
-96dB
-102dB
-108dB
-114dB
0°
-20°
-40°
-60°
-80°
-100°
-120°
-140°
-160°
-180°
-200°
-220°
-240°
-260°
1Hz
10Hz
100Hz
1KHz
10KHz
100KHz
1MHz
10MHz

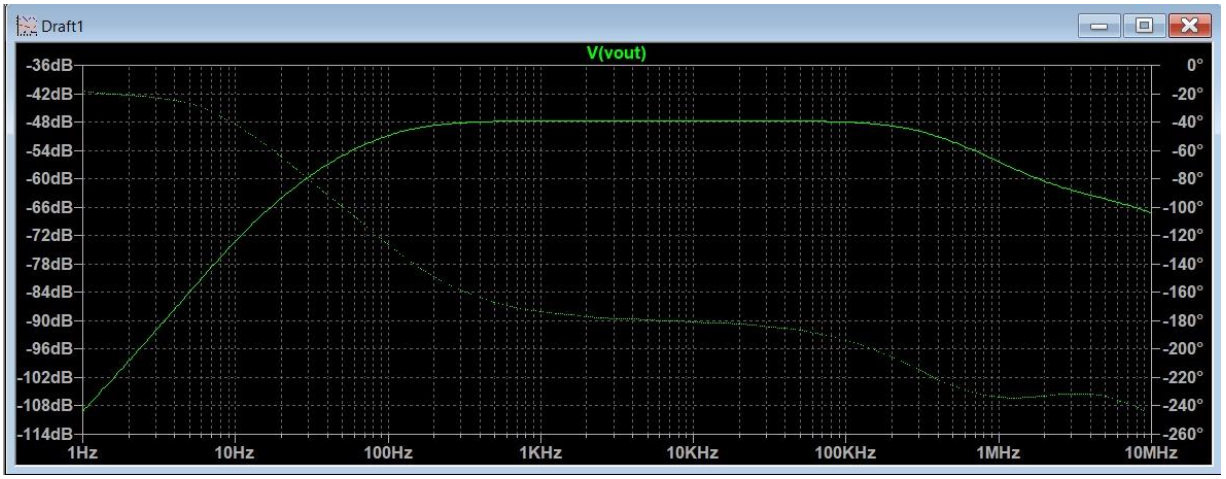
1M Hz


Agilent Technologies
DSO1012A
Oscilloscope
100MHz
2GSa/s
Menu
On/Off
STOP
20.00ns/
AUTO
Vmax=-34.0mV
Vmin=-76.0mV
Vpp=42.0mV
Vtop=-34.0mV
Vbas=-76.0mV
Vamp=42.0mV
Vavg=-53.5mV
Vrms=54.4mV
Vovr=19.0%
Vpre=33.3%
Prd=10.0ns
Freq=100MHz
Rise=73.2ns
Fall=*****
+Wid=4.00ns
-Wid=6.00ns
+Duty=40.0%
-Duty=60.0%
Undo
Prd=10.0ns
50.0mV/
Push
to Select
Cursors
Auto-
Scale
Vertical
Push
Math
Push
to Zero


Draft1
V(vout)
-36dB
-42dB
-48dB
-54dB
-60dB
-66dB
-72dB
-78dB
-84dB
-90dB
-96dB
-102dB
-108dB
-114dB
0°
-20°
-40°
-60°
-80°
-100°
-120°
-140°
-160°
-180°
-200°
-220°
-240°
-260°
1Hz
10Hz
100Hz
1KHz
10KHz
100KHz
1MHz
10MHz

10M Hz





19. Compare the calculated (step 1, 3), simulated (step 2, 4) and measured (step 10, 16) responses.

Answer:

Our values which we measure on ltspice and the values was measure on lab are close but our measured values in lab are a little bit different from others. We built our both two circuits correctly and measure the values and We didnt think we have a trouble with ltspice. Because of our calculations and simulation measurements are close, we think maybe we had a mistake when we were measuring in labratory. Maybe it is about the cables, breadbords or ac source.